

Building Quantum Futures in Africa: Curriculum Development, Challenges, and Opportunities in Experimental Physics Education

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Quantum Science and Technology Across Africa Workshop
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Why develop quantum sciences and technology in Africa ?

- ▶ Global quantum revolution : Quantum tech (computing, sensing, communication) is transforming industries.
- ▶ Africa risks becoming a consumer, not a contributor, without proactive engagement.
- ▶ Africa's potential : building local expertise.
- ▶ Economic & scientific impact : Quantum research can drive innovation in materials, energy, AI, and cybersecurity.

What happens if Africa misses the quantum wave?

- ▶ Dependence on foreign tech providers.
- ▶ Widening digital inequalities.
- ▶ Missed opportunity for job creation and innovation.

Call to action :

- ▶ Establish quantum education programs.
- ▶ Fund research infrastructure and startups.
- ▶ Join international quantum collaborations.

Outline

1- Quantum Physics Education in Africa

2- Quantum Sciences in Morocco

3- Proposed Master's Program in Experimental Quantum Physics

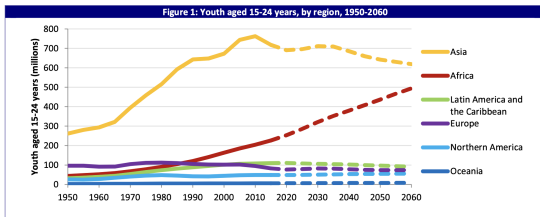
4- Conclusion

1- Quantum Physics Education in Africa

- ▶ Study of physical systems at atomic and subatomic levels.
- ▶ Includes quantum computing, quantum communication, and quantum sensing.
- ▶ Foundation for next-generation technologies.

Statistics & Progress

- ▶ Africa's demographic advantage :



Data source: United Nations (2012) World Population Prospects: The 2012 Revision

- ▶ Over 500 million young people under 25 today.
- ▶ Investing in quantum creates high-tech jobs, curbs "brain drain".
- ▶ Workshops/Conferences : African School of Physics, Quantum Africa Conference...

Current state of quantum education in Africa :

- ▶ Few universities offer formal quantum programs.
- ▶ Most programs are theoretical ; few experimental labs.
- ▶ Barriers : Equipment costs, faculty training, and policy support.

Challenges in quantum education :

- ▶ Lack of trained educators in quantum science.
- ▶ Limited lab infrastructure for experimental training.
- ▶ Minimal integration into national curricula.
- ▶ Limited industry-academia partnerships.
- ▶ Fragmented efforts ; lack of coordinated funding.

2- Quantum Sciences in Morocco

Statistics & Progress

- ▶ Students & Programs : Number of universities offering quantum-related courses (e.g., Mohamed V University, Abdelmalek Essaadi University).
- ▶ Laboratories & Teams : Existing research groups (e.g., LPT, R&DIS, UMP6...) and their focus areas (quantum optics, QKD).
- ▶ Projects & Funding : National/international collaborations (e.g., with CNRST/OCP, EU).
- ▶ Conferences/Workshops : Events such as "Quantum Africa", "Moroccan Quantum Days", "Quantum Morocco", "School/Workshop"....

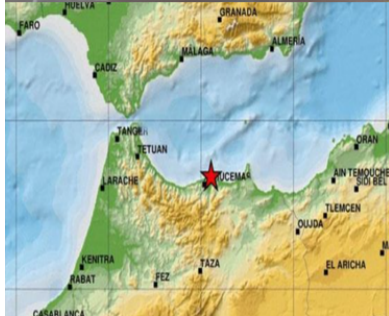
Challenges :

- ▶ Limited lab infrastructure for hands-on quantum experiments.
- ▶ Need for standardized curricula linking theory to practice.
- ▶ Brain drain : 40% of graduates leave for PhDs abroad (Morocco $\rightarrow$ EU/US).
- ▶ Low funding for quantum-specific programs.
- ▶ Need for international exposure & collaboration.
- ▶ No dedicated experimental quantum physics program.
- ▶ Few courses on quantum engineering/applications.
- ▶ Student demand : a little interest in quantum careers.
- ▶ Few specialized faculty in quantum computing/information.
- ▶ Insufficient industry-academic linkage.

Growth opportunities :

- ▶ Introduce specialized undergraduate/graduate quantum programs.
- ▶ Develop online quantum courses in English/Arabic/French.
- ▶ Partnerships between African universities and institutions.
- ▶ Partner European institutions (Erasmus+,...)
- ▶ Invest in lab kits, simulators, and quantum sandbox platforms.

3- Proposed Master's Program in Experimental Quantum Physics



School and Workshop on Quantum Information Theory and Thermodynamics at the Nanoscale



24 February - 6 March 2020
Al Hoceima, Morocco

Further information:
School webpage: <http://indico.ictp.it/event/16023/>
Workshop webpage: <http://indico.ictp.it/event/16106/>
enr300@ictp.it

This School (24 - 28 February) and Workshop (2 - 6 March) focus on the forefront of the field of quantum technologies, a fast growing research area involving questions of both fundamental and practical relevance.

What are the roles and value of coherence and entanglement in solid state devices in quantum information, and in the working and efficiency of mesoscopic quantum engines?
What are the mechanisms behind thermalization of closed quantum systems and what is the role of non-Markovianity in the physics of quantum information?

These topics will be covered during the School, with a subsequent Workshop designed for graduate students and junior researchers, to offer quality training and the opportunity to meet some leading experts in the field.

Topics:

- Open Quantum Systems
- Non-equilibrium Quantum Dynamics
- Complex Quantum Systems
- Quantum Machine Learning
- Many-body Quantum Systems
- Physical Foundations of Quantum Information

How to apply:

School webpage:
<http://indico.ictp.it/event/16023/>
Workshop webpage:
<http://indico.ictp.it/event/16106/>
To participate in both activities it is necessary to submit two distinct applications.
Resume available are particularly encouraged to apply.

Grants:

A limited number of grants are available to support the attendance of selected participants, with priority given to participants from developing countries. There is no registration fee.

Directors:

A. JAFFRÈ (Institut Henri-Poincaré, Grenchen)
A. BERRIUTI (University of Trieste)
A. EL ALAÏ (University Abdelmalek Beaudou, Al Hoceima)
M. EL BAZ (University Mohammed V, Rabat)
A. EL HADJER (University Abdelmalek Beaudou, Al Hoceima)
Y. KUSUNO (University Mohammed V, Rabat)
T. MOURABIT (University Abdelmalek Beaudou, Al Hoceima)
M. NAJIA (University of Palermo)
M. NERENBERG (Queen's University, Belfast)
A. SCARABINO (ICTP Trieste)
Y. TABALAT (University Mohammed V, Rabat)
L.R. BREGOLI (University Mohammed V, Rabat)

Lecturers:

G. BENNETT (University of Innsbruck, China)
A. BECHTEREF (Albert-Ludwigs University of Freiburg)
D. CAMPBELL (University College Dublin)
D. CHEUSCHIN (Nicolaus Copernicus University of Torun)
L. BRUGES (University of Bologna)
M. REIS (University of Milano)

Workshop Speakers:

M. ADDOZI (International University of Rabat)
M. BERRIUTI (University Mohammed V, Rabat)
K. BERTHE (Delft University of Technology)
D. BOCHINE (Université Paul Sabatier Toulouse)
H. EL BOUCCH (Mohammed I University, Oujda)
D. BRAN (University of Tübingen)
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G. MONTI (University of Southampton)
J. PROLA (Aalto University)
S. RABELO (University of Bath)
R. WHITNEY (EPANIC CNRS Grenchen)
F. ZAMBONI (Comenius University Bratislava)
F. ZUCCHETTI (Universität Wien, Austria)
* to be confirmed

ICTP Scientific Contact:

A. SCARABINO

Deadlines:

10 November 2019

24 November 2019

For applications and requesting financial support, not needing visa





Master in Quantum Information : Modeling and Applications



- ▶ First academic year : 2021/2022
- ▶ Coordinator : Abderrahim El Allati
- ▶ 4 generations.
- ▶ Master IQMA focused on the physical sciences without neglecting the contribution of engineering sciences. This can allow students to start their PhD studies easily.

Descriptions of 1st year Master modules

Semestre	Module code	Module title
S1	M11	Scientific English & PYTHON Programming
	M12	Mathematics for physics
	M13	Functional analysis
	M14	Physical properties of materials
	M15	Radiation matter interaction
	M16	Advanced quantum mechanics
S2	M21	Semiconductor Materials
	M22	English for science and entrepreneurship
	M23	Quantum information theory
	M24	Plasma physics
	M25	Nanotechnology for photovoltaic cells
	M26	Numerical modeling and simulation

Descriptions of 2nd year Master modules

Semestre	Module code	Module title
S3	M31	Semiconductor physics and applications
	M32	Quantum Technologies
	M33	Quantum optics
	M34	Nanostructured Photonic Crystals
	M35	Communication and quantum cryptography
	M36	Quantum Decoherence
S4	M37	PFE
	M38	
	M39	
	M40	
	M41	

Master's thesis

- ▶ Quantum thermometry (Robert Withiny)
- ▶ Quantum machine learning (Lucas Gian)
- ▶ Quantum thermodynamics (quantum engines, quantum refrigerators, quantum batteries)-(Felip Bara, OE Mustecaplioglu)
- ▶ Open quantum systems (Prof. Bassano Vacchini)
- ▶ Quantum optics (Prof. S Al-Kuwari)
- ▶ Quantum optomechanics (IP Castillo, F. Saif)
- ▶ Quantum estimation (Prof. Nasser Metwally)
- ▶ Quantum Discord (Akram Touil)

List of publications

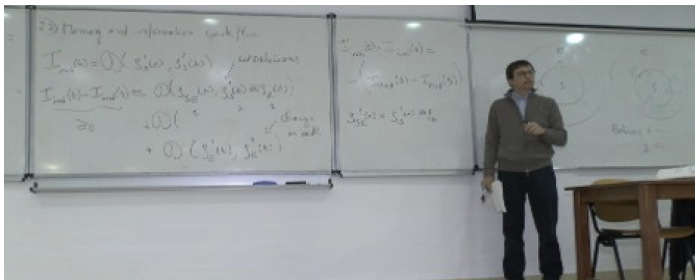
1. Y. Aiache, A. Ullah, OE. Mustecaplioglu, A. El Allati, **Phys Rev. A** **111** (6), **062619** (2025)
2. H. Assil, A. El Allati and G L Giorgi, **Phys Rev. A** **111** (2), **022412** (2025)
3. A. Khoudiri, A. El Allati, OE. Mustecaplioglu, and K. El Anouz, **Phys Rev. E** **111** (4), **044124** (2025)
4. Y. Aiache, C. Seida, K. El Anouz and A. El Allati, **Physics Letters A** **496**, **129320** (2024)
5. Y. Aiache, A. El Allati, K. El Anouz, S. Al-Kuwari, **J. Phys. A : Math. Theor.** **57** **195303** (2024)
6. Y. Aiache, A. El Allati, and K. El Anouz, **Phys. Rev. A** **110** (3) (2024)
7. Y. Aiache, C. Seida, K. El Anouz and A. El Allati, **Phys. Rev. E** **110**, **024132** (2024)
8. Z. Imara, K. El Anouz and A. El Allati, **J. Phys. B : At. Mol. Opt. Phys.** **57** (2024) **185501**
9. M. Ait Bahadou, A. El Allati, K. El Anouz, **Physica Scripta** (2024)
10. Z Imara, K El Anouz, F Saif, A El Allati, **J. Phys. B : At. Mol. Opt. Phys.** **57** **235501**
11. Y. Aiache, K. El Anouz, N. Metwally and A. El Allati, **Phys. Rev. E** **109**, **034101** (2024)

- Quantum Information Club

- Book (in preparation) : From classical electronics to Q-AI :

Technologies and Applications

Lectures by Prof. B. Vacchini Non-Markovian open quantum systems dynamics, February 2023





SCIENTIFIC DAY IN QUANTUM INFORMATION CELEBRATING 100 YEARS OF QUANTUM MECHANICS

10 July 2025

Faculty of Sciences and Techniques Al-Hoceima

This day is a moment of privileged exchanges and collective reflection around the several challenges in quantum information in the occasion of 100 years of quantum mechanics for which the Faculty of Sciences and Techniques Al-Hoceima invites its members and other speakers from different Moroccan faculties to mobilize through its new strategy to develop quantum technologies in Morocco. Indeed, the main objectives of this event relate to the development of scientific research in quantum information within the Faculty of Sciences and Techniques Al-Hoceima.

Honor Committee

Pr. Bouchta EL MOUMNI : President of Abdelmalek Essâadi University

Pr. Mohammed BAKKALI : Dean of the Faculty of Sciences and Techniques Al-Hoceima

Pr. Hinde CHERKAOUI DEKKAKI : Vice-President of Abdelmalek Essâadi University in charge of Scientific Research and Cooperation

Organizing Committee

Pr. Abderrahim El Allati (FSTH, UAE, Morocco)

Pr. Khadija El Anouz (FSTH, UAE, Morocco)

Keynote Speakers

Pr. Mohammed Daoud (Ibn Tofail University)

Pr. Nabil Habiballah (Ibn Zohr University)

Pr. Khadija El Anouz (FSTH, UAE, Morocco)

International Workshop



Strategic Empowerment of Graduate Research : Integrating Advanced Artificial Intelligent and Machine Learning Techniques in Academic Inquiry

December 09-24, 2024, FSTH, Al-Hoceima Morocco

This workshop is organized by the Faculty of Sciences and Techniques of Al-Hoceima, Morocco in collaboration with the National Center for Scientific and Technical Research within the framework of the FINCOME project.

This international workshop presents a 16-day intensive training program aimed at substantially enhancing Artificial Intelligent and machine learning skills among Master's, PhD students and PhD researchers.

Director

Abderrahim EL ALLATI (FSTH, UAE, Morocco)

Contact: eabderrahim@uae.ac.ma

Expert

Ismael EL MOUDDEN

Director, AI Strategic Integration and Healthcare Innovation, Faculty Affairs and Professional Development

Proposed Master's Program

Program Vision

- ▶ Combines theory with hands-on labs (optics, cryogenics, QKD).
- ▶ Collaborations : Partner with African Union / EU institutions for curriculum design and equipment.
- ▶ Modules :
 - ▶ Quantum Optics (Photonics Lab).
 - ▶ Cryogenics (Superconducting Qubits).
 - ▶ QKD (Cybersecurity Apps).

A significant challenge we face is a lack of experienced experimentalists and established laboratory infrastructure in this area. We are particularly interested in individuals or groups who could :

- ▶ Share their expertise in experimental quantum physics.
- ▶ Guide on setting up and equipping quantum laboratories.
- ▶ Offer opportunities for student internships or research exchanges.
- ▶ Collaborate on developing new experimental modules and teaching materials.
- ▶ Internship for Master's students.
- ▶ Next year, we will have a new accreditation with the title **"Quantum Science and Technologies"**, as we aim to develop primarily the experimental side.

Aims :

- ▶ MSc in Experimental Quantum Physics.
- ▶ Bridge theory-practice gap with hands-on labs.
- ▶ Enroll students from African countries.
- ▶ Give courses in English.
- ▶ Participate in teaching.
- ▶ Supervisor students.

Curriculum structure content :

- ▶ Year 1 : Core courses (QM, QIT).
- ▶ Year 2 : Lab rotations + industry internship.

Master in Quantum Sciences and Technologies (QST)

Semestre 1 :

- ▶ Advanced Quantum Mechanics
- ▶ Physical Properties of Materials
- ▶ Linear Algebra and Probability Theory
- ▶ Quantum Programming
- ▶ Quantum Optimization and Control
- ▶ Functional Analysis

Semestre 2 :

- ▶ Quantum Communication and Cryptography
- ▶ Quantum Device Physics
- ▶ Quantum Information Theory
- ▶ Semiconductor Materials
- ▶ Topological Qubits : Foundations and Applications
- ▶ Quantum Machine Learning

Semestre 3 :

- ▶ Superconducting Circuits and Quantum Platforms
- ▶ Nanostructured Photonic Crystals
- ▶ Quantum Optics and Information Technologies
- ▶ Open Quantum Systems
- ▶ Quantum Thermodynamics

Semestre 4 : Master's thesis

Our vision for this new program includes modules such as "Practical Work : Laboratory Experiments," covering topics like :

- ▶ Manipulation of superconducting or photonic qubits.
- ▶ Implementation of quantum communication protocols.
- ▶ Study of optomechanical systems in the laboratory.
- ▶ Cold atoms, ion traps, and superconducting qubits.
- ▶ Quantum photonics and single-photon sources.
- ▶ Quantum materials (topological insulators, superconductors).

We need your expertise. Can you help us :

- ▶ Design lab modules ?
- ▶ Host student exchanges ?
- ▶ Donate equipment ?

Together, we can bridge the gap between theory and practice. Let's turn "brain drain" into "brain gain".

4- Conclusion

- ▶ Urgency : Africa must act now to avoid missing the quantum revolution.
- ▶ Governments : Fund infrastructure and research.
- ▶ Universities : Adopt standardized quantum curricula.
- ▶ Industry : Invest in R&D partnerships.
- ▶ The quantum age has arrived. Africa has a choice : act or be left behind.

"Quantum Futures Start Today
Let's Build Them Together"
Thank you for your attention